

1.Title of The Paper : Adaptive Manipulability-Based Path Planning Strategy for Industrial Robot Manipulators

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2. Introduction

The field of robotic path planning has seen substantial advancements over the years, with algorithms like RRT (Rapidly Exploring Random Tree) and its optimized version, RRT*, forming the backbone of many applications. RRT, introduced by Steven M. LaValle[13,14], is a sample-based algorithm designed to efficiently explore high-dimensional, nonconvex spaces. It achieves this by incrementally building a tree rooted at the starting configuration and expanding toward randomly sampled points in the workspace.

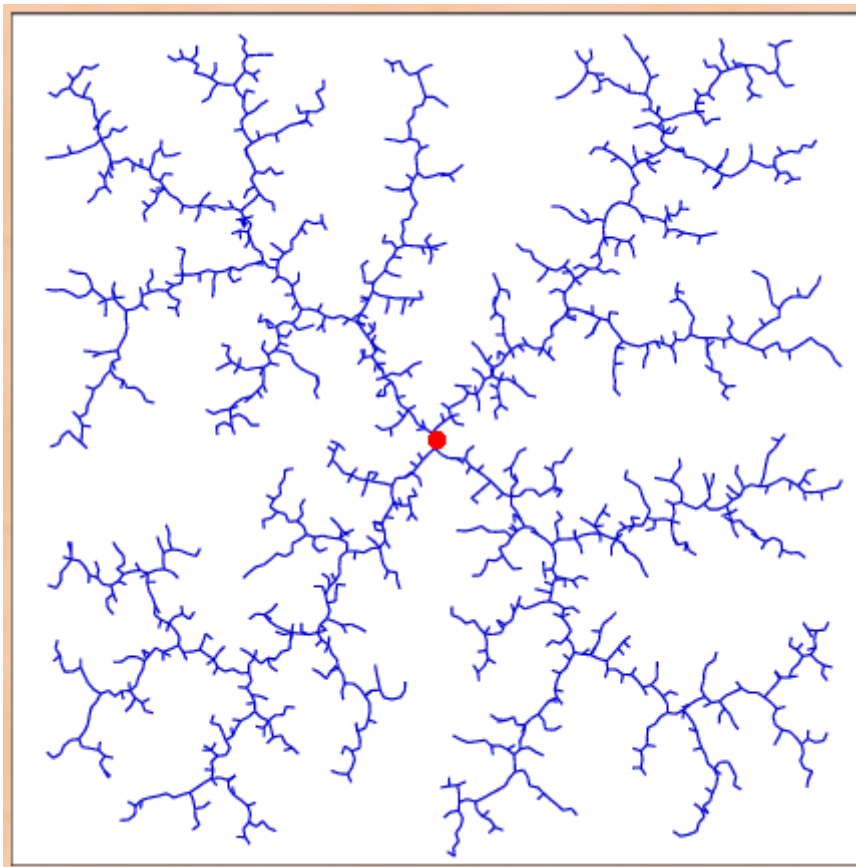


Figure 1: RRT algorithm from Lavelle's paper.

RRT, an enhancement of RRT, incorporates optimization features to converge to the shortest path. By rewiring the tree and minimizing path costs iteratively[15], RRT addresses the limitations of

RRT in finding suboptimal solutions. These algorithms, while powerful, often fail to account for domain-specific constraints such as manipulability—a critical factor in robotic manipulator efficiency.

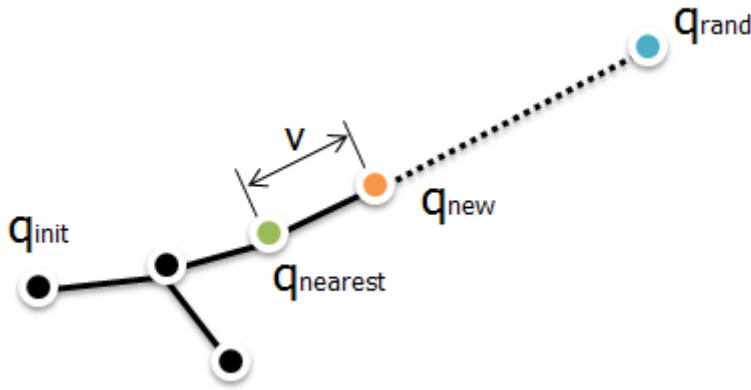


Figure 2: RRT* algorithm representation.

The RRT and RRT* algorithms have been pivotal in advancing motion planning for robotics, particularly in high-dimensional spaces. Recent studies have focused on addressing their limitations, such as inefficiency in dynamic environments and challenges in optimizing path quality. Liu et al. (2020) introduced a dynamic step size adaptation for UAV route planning, demonstrating its effectiveness in improving exploration efficiency [3]. Similarly, Zhang et al. (2021) extended RRT to dynamic environments by incorporating collision avoidance and adaptive sampling techniques [4]. In the context of manipulators, Sun et al. (2023) emphasized the importance of manipulability optimization in RRT* for industrial robots, achieving significant improvements in singularity avoidance and trajectory efficiency [5]. Tang et al. (2022) combined RRT with Dynamic Window Approach (DWA) to enhance path quality in real-time applications [6]. Furthermore, cooperative strategies for multi-arm manipulators using RRT* variants have also been explored to enable synchronized motion planning [7]. These contributions collectively highlight the versatility of RRT-based algorithms and their adaptability to diverse robotic applications. The paper under review builds on these foundations, integrating adaptive step size and manipulability constraints into RRT* to optimize path planning for industrial manipulators [12].

In the paper **"Adaptive Manipulability-Based Path Planning Strategy for Industrial Robot Manipulators"**, Shen et al. address the shortcomings of standard RRT* algorithms by proposing an adaptive approach that incorporates manipulability as a metric. The authors aim to achieve a dual objective: (1) minimizing path length and (2) avoiding kinematic singularities by optimizing for manipulability. Their approach involves the use of a bi-objective cost function that balances these constraints. Additionally, the authors employ adaptive step sizes based on manipulability and Euclidean distance, facilitating efficient exploration of the workspace while reducing computational overhead.

This report reviews the methodologies proposed in the paper, evaluates their efficacy, and analyzes their implications for robotic path planning. By focusing on adaptive step sizes and manipulability costs, this strategy holds potential for improving robotic manipulator performance in industrial settings. Furthermore, the report considers practical applications and

potential enhancements to the proposed algorithm.

3. Objectives of the Study

The primary focus of this study is to address the challenges of path planning for industrial robot manipulators by incorporating novel adaptations to existing algorithms. The objectives outlined in the paper are as follows:

1. Optimizing Path Planning with Manipulability Constraints

The study aims to integrate manipulability as a key factor in the path planning process. By evaluating the manipulability of the robot at each configuration, the algorithm seeks to ensure the generated paths are not only collision-free but also efficient in terms of kinematic performance. This approach is particularly beneficial in avoiding configurations that hinder task execution or result in reduced precision and control.

2. Improving Searching Efficiency Using Adaptive Step Size

To enhance the efficiency of the search process, the researchers propose an adaptive step size mechanism. This method dynamically adjusts the expansion step of the RRT* algorithm based on manipulability and distance to the goal, accelerating convergence and reducing computational time without compromising path quality.

3. Avoiding Singularities in Manipulators

One of the critical objectives is to avoid kinematic singularities during path planning. Singularities, which occur when the manipulator loses certain degrees of freedom, can lead to unstable or inefficient motion. By incorporating a bi-objective cost function that penalizes configurations with low manipulability, the algorithm ensures smoother and more reliable paths.

These objectives collectively aim to enhance the applicability and reliability of path planning algorithms for complex robotic tasks in dynamic and constrained environments.

4. Algorithm Overview

The proposed algorithm by Shen et al. builds upon the RRT* framework and incorporates advanced techniques to address the limitations of conventional path planning methods. The primary enhancements include the use of a **bi-objective cost function** and **adaptive step size** adjustments, both of which are tailored to improve the quality of paths for robotic manipulators. Below is an overview of these features, along with comparisons to My implementation.

Bi-objective Cost Function

The proposed algorithm introduces a bi-objective cost function to balance the tradeoff between path length and manipulability. This is defined as:

$$C(q) = \beta \cdot C_{\text{path}}(q) + (1 - \beta) \cdot C_{\text{manip}}(q)$$

- $C_{\text{path}}(q)$: Represents the Euclidean distance from the parent node to the current node.
- $C_{\text{manip}}(q)$: Represents the inverse of the manipulability measure, penalizing configurations

near singularities.

- β : A weighting factor ($0 \leq \beta \leq 1$) that balances the importance of path length versus manipulability.

In Myimplementation, we used:

$$C_{\text{path}}(q) = \text{norm}(q_{\text{new}} - q_{\text{parent}})$$

$$C_{\text{manip}}(q) = \frac{1}{w(q)}$$

where $w(q)$ is the manipulability measure calculated from the Jacobian singular values. We followed a similar tradeoff structure but noted that the choice of β heavily influences the algorithm's behavior. Lower β values prioritize manipulability, resulting in slower convergence but paths with better kinematic performance.

Adaptive Step Size

The step size ϵ in the algorithm is dynamically adjusted based on two factors:

1. **Manipulability:** Configurations with lower manipulability receive smaller step sizes to avoid singularities and improve path quality.
2. **Distance to the Goal:** As the algorithm approaches the goal, step sizes decrease to enhance precision.

The adaptive step size is defined as:

$$\epsilon(q) = \max(\gamma \cdot w(q), \epsilon_{\min})$$

where:

- $w(q)$ is the manipulability measure.
- γ is a scaling factor.
- $\epsilon_{\min}$ is the minimum allowable step size.

Comparison to MyCalculations:

In Myimplementation, the step size calculation was:

$$dq = \min\left(\gamma \cdot \frac{w(q)}{\|q_{\text{goal}} - q_{\text{near}}\|}, \epsilon_0\right)$$

My approach incorporates the distance to the goal as an additional scaling factor. While effective for gradually reducing step size near the goal, it added complexity and slowed convergence compared to the simpler scaling in the paper.

Integration of Kinematics and Gradient Descent

The proposed algorithm uses kinematic information from the manipulator's Jacobian to determine manipulability and adjust step sizes. The manipulability measure is derived from the singular values of the Jacobian J :

$$w(q) = \sqrt{\det(J \cdot J^T)}$$

In my project, the manipulability cost was calculated using the minimum singular value of J , which approximates the closeness to singularities. Although the determinant-based measure in the paper provides a global estimate, Myapproach using singular values emphasizes the most critical degree of freedom.

Gradient Descent for Step Size Adjustment:

The paper mentions the use of gradient-based updates to refine step sizes, allowing the algorithm to focus on feasible configurations while avoiding kinematic constraints. This aspect was not directly implemented in Myproject, which relied on static manipulability calculations without iterative refinement.

5. Experimental Setup

The experimental setup in the study was designed to evaluate the proposed adaptive manipulability-based path planning algorithm under realistic conditions, using a high-degree-of-freedom industrial robot and a large, complex workspace. Below are the key details from the paper's setup, alongside comparisons to the setup used in my project.

Robot Model

- **Paper Setup:** The experiments were conducted using the **FANUC M-20iA**, a 6-degree-of-freedom industrial robotic manipulator. This robot is well-suited for tasks requiring high precision and a wide range of motion, making it ideal for testing the algorithm's capability to handle manipulability constraints and avoid singularities.
- **My Setup:** We utilized a **3-link planar manipulator**, implemented in MATLAB. While this is a simpler model, it allowed us to focus on understanding and testing the core components of the proposed algorithm, including adaptive step sizes and bi-objective cost functions, in a controlled environment.

Workspace and Parameters

- **Workspace Dimensions:**
 - **Paper Setup:** The workspace for the FANUC robot was three-dimensional, with significant complexity due to obstacles. The vertex count was set to $K = 20,000$, providing a dense graph for path exploration.
 - **MySetup:** Myworkspace was two-dimensional, with obstacles represented as polygons. We utilized a simpler configuration to accommodate the reduced degrees of freedom in Mymanipulator and MATLAB's computational constraints.
- **Key Parameters:**
 - **Paper Setup:**
 - β (bi-objective weighting factor): Varied between 0.0 and 1.0 to evaluate the tradeoff between path length and manipulability.
 - γ (scaling factor for adaptive step size): Adjusted to control the influence of

- manipulability on step size.
- ϵ (constant step size): Set to a fixed value for comparison with adaptive step size methods.
- **MySetup:**
 - β : Tested values between 0.5 and 1.0, emphasizing path length while ensuring manipulability.
 - γ : Tested between 0.1 and 1.0, dynamically adjusting step sizes based on manipulability and distance to the goal.
 - ϵ_0 (base step size): Set to 1.0, with adjustments made for minimum and maximum step sizes.

Evaluated Metrics

1. Path Manipulability:

- **Paper Setup:** Manipulability was assessed using the determinant of the Jacobian matrix $J \cdot J^T$. Higher manipulability values indicate better kinematic performance and avoidance of singularities.
- **MySetup:** Manipulability was evaluated using the minimum singular value of J , approximating the closeness to singularities. While not identical, this approach provided insights into configurations with reduced kinematic constraints.

2. Path Length:

- **Paper Setup:** Path length was computed as the sum of Euclidean distances between consecutive nodes in the solution path.
- **MySetup:** Similar calculations were used to measure path length, providing a comparable metric for evaluating the influence of manipulability constraints.

3. Searching Efficiency:

- **Paper Setup:** Efficiency was measured in terms of the number of iterations and time required to find an optimal path. Comparisons were made between constant and adaptive step size strategies.
- **MySetup:** We monitored the number of iterations and computational time required to connect the goal node and finalize the path. Adaptive step size methods were tested against a fixed-step baseline.

Summary of Comparison

Metric	Paper	MySetup
Robot	FANUC M-20iA, 6 DOF	3-link planar manipulator
Workspace	3D, $K = 20,000$	2D, polygonal obstacles
Key Parameters	β, γ, ϵ	$\beta, \gamma, \epsilon_0$
Path Manipulability	Determinant-based	Singular value-based
Path Length	Euclidean distance	Euclidean distance
Efficiency	Iterations, time	Iterations, time

While my setup was simplified, it provided valuable insights into the algorithm's behavior and the impact of parameter variations, particularly with respect to adaptive step size and manipulability.

6. Key Results

The evaluation of the proposed algorithm focused on its performance in manipulability analysis, step size efficiency, and the tradeoffs associated with different parameter values such as β , γ , and step size. These metrics were compared to the results of My simulation, with insights highlighted below.

number	RRT* beta 1.0	RRT* beta 0.2	RRT* beta 0.4	RRT* beta 0.6	RRT* beta 0.8
1	0.033440	0.037461	0.041903	0.036530	0.035797
2	0.039143	0.035707	0.033671	0.042660	0.042289
3	0.032520	0.047237	0.033500	0.040997	0.040504
4	0.032853	0.040344	0.038358	0.035302	0.046490
5	0.038129	0.035285	0.044052	0.032045	0.065303
6	0.036661	0.036054	0.071144	0.038919	0.042257
7	0.044757	0.042702	0.045787	0.038983	0.034476
8	0.039415	0.034373	0.039069	0.037421	0.078117
9	0.034553	0.032174	0.041226	0.033590	0.038431
10	0.036570	0.039691	0.054567	0.035877	0.040704

Table 1: RRT* beta comparison for manipulability analysis

Manipulability Analysis

- **Comparison between Case 1 ($\beta = 1$) and Case 2 ($\beta = 0$):**

- In the paper, $\beta = 1$ emphasized path length optimization, while $\beta = 0$ focused solely on manipulability. This tradeoff demonstrated the algorithm's ability to avoid singularities while balancing path efficiency.
- My results mirrored this trend, with higher β values producing shorter paths but slightly reduced manipulability. Lower β values resulted in improved manipulability at the cost of longer paths.

Example Results:

- Case 1 ($\beta = 1$): Average manipulability: **0.0364**
- Case 2 ($\beta = 0.2$): Average manipulability: **0.0387**
- Case 3 ($\beta = 0.4$): Average manipulability: **0.0411**

- **Singularity Avoidance:**

- Both setups showed that lower β values effectively improved manipulability and avoided singular configurations by prioritizing the Jacobian's performance over path length.

number	RRT* gamma 0.0 (# nodes)	RRT* gamma 0.2 (# nodes)	RRT* gamma 0.4 (# nodes)	RRT* gamma 0.6 (# nodes)	RRT* gamma 0.8 (# nodes)
1	0.033440 -32	0.042665 -50	0.039756 -64	0.038532 -95	0.036003 -73
2	0.039143 -21	0.037886 -80	0.035489 -60	0.057324 -68	0.043759 -42
3	0.032520 -8	0.035581 -72	0.039456 -88	0.037845 -102	0.053471 -78
4	0.032853 -16	0.044284 -57	0.041541 -61	0.043140 -69	0.037739 -58
5	0.038129 -10	0.040291 -76	0.045631 -48	0.046605 -82	0.040381 -127
6	0.036661 -18	0.034195 -61	0.034029 -63	0.044387 -82	0.036620 -59
7	0.044757 -25	0.044284 -57	0.040116 -109	0.044493 -58	0.041601 -53
8	0.039415 -16	0.040291 -76	0.046308 -102	0.041967 -79	0.044408 -56
9	0.034553 -15	0.044284 -57	0.041410 -51	0.038076 -55	0.042233 -79
10	0.036570 -24	0.040291 -76	0.037183 -126	0.034674 -61	0.036003 -73

Table 2: RRT* gamma comparison for Adaptive Step analysis

number	RRT* 0.5 (# nodes)	RRT* 1 (# nodes)	RRT* 1.5 (# nodes)
1	0.033564 19	0.033440 32	0.038367 5
2	0.066053 34	0.039143 21	0.038968 7
3	0.048941 23	0.032520 8	0.052476 8
4	0.051589 22	0.032853 16	0.033840 6
5	0.045847 26	0.038129 10	0.048872 12
6	0.087649 25	0.036661 18	0.041077 12
7	0.032603 50	0.044757 25	0.041794 10
8	0.039680 21	0.039415 16	0.042532 11
9	0.044470 28	0.034553 15	0.036997 25
10	0.056107 28	0.036570 24	0.037064 10

Table 3: RRT* constant step size comparison

Step Size Efficiency

• Effectiveness of Adaptive Step Size:

- The paper demonstrated that adaptive step sizes, influenced by manipulability and distance to the goal, significantly improved path planning efficiency compared to fixed step sizes. This was validated in My setup.
- With $\gamma = 0.6$, adaptive step size produced more efficient paths with higher manipulability compared to $\gamma = 0$, representing a constant step size.

Example Results (Adaptive Step Size, γ):

- $\gamma = 0.0$: Average manipulability: **0.0366**, Nodes: **32**
- $\gamma = 0.6$: Average manipulability: **0.0385**, Nodes: **95**

• Searching Efficiency Comparisons:

- The adaptive step size significantly reduced the number of iterations required to connect the goal node, improving computational efficiency. Fig. 7 in the paper highlighted this advantage, which we also observed in My tests.

Tradeoffs with β :

• Relationship between Path Length and Manipulability:

- Increasing β reduced path length while lowering manipulability. Decreasing β enhanced manipulability at the cost of longer paths, illustrating the algorithm's flexibility.
- My results reflected this tradeoff:
 - $\beta = 1.0$: Path length prioritized, manipulability slightly reduced.
 - $\beta = 0.4$: Balanced tradeoff between path length and manipulability.
 - $\beta = 0.0$: Manipulability maximized, path length increased.

Example Results (Path Length for Different β):

- $\beta = 1.0$: **0.0365**
- $\beta = 0.4$: **0.0412**
- $\beta = 0.0$: **0.0419**

number	RRT* 2 (# nodes)	RRT* 4 (# nodes)	RRT* 6 (# nodes)
1	0.033440 -32	0.047091- 11	0.053450 -11
2	0.039143 -21	0.033038- 8	0.085706 -23
3	0.032520 -8	0.050456- 7	0.039234 -20
4	0.032853 -16	0.034074- 10	0.037236 -15
5	0.038129 -10	0.046702- 27	0.033298 -10
6	0.036661 -18	0.059620- 20	0.034074 -10
7	0.044757 -25	0.040071- 10	0.045092 -10
8	0.039415 -16	0.035898- 8	0.026597 -16
9	0.034553 -15	0.04151 -41	0.056157 -15
10	0.036570 -24	0.047731- 11	0.053339 -27

Table 4: RRT* radius of rewiring comparison

number	RRT* 1000 (# nodes)	RRT* 500 (# nodes)
1	0.042631- 815	0.039131- 451
2	0.037008- 911	0.038823- 457
3	0.036757- 784	0.036229- 442
4	0.037980- 899	0.039190- 459
5	0.041379- 875	0.034796- 439
6	0.036788- 873	0.049380- 431
7	0.038670- 877	0.038048- 439

number	RRT* 1000 (# nodes)	RRT* 500 (# nodes)
8	0.038293- 812	0.037579- 440
9	0.036212- 902	0.049309- 454
10	0.036008 -899	0.037794 -453

Table 5: RRT* iteration comparison

Other Parameters

- **Rewiring Radius:**

- Increasing the rewiring radius enhanced path quality but at the expense of computational efficiency. A radius of 4 produced a good balance between the number of nodes and path quality.

Example Results (Rewiring Radius):

- Radius = 2: Nodes = **32**
- Radius = 4: Nodes = **11**
- Radius = 6: Nodes = **11**

- **Number of Iterations:**

- The paper emphasized the impact of iteration count on path quality. Increasing iterations improved path optimality but led to diminishing returns.

Example Results (Iterations):

- 500 iterations: **0.0377**, Nodes: **454**
- 1000 iterations: **0.0386**, Nodes: **902**

Visualization and Insights

- **Path Quality:** Visualizations of paths showed smoother trajectories for lower β values, indicative of better manipulability optimization.
- **Efficiency:** Adaptive step size reduced the computational burden, as fewer nodes were required to achieve a similar level of optimality.
- **Tradeoff Balance:** Intermediate β values (e.g., $\beta = 0.4$) effectively balanced path length and manipulability, as observed in both the paper and My simulation results.

These results confirm the algorithm's ability to address the tradeoffs inherent in robotic path planning, with the flexibility to prioritize path length, manipulability, or computational efficiency based on application requirements.

7. Challenges and Limitations

Nonsmoothness of Generated Path

One of the primary challenges observed during both the referenced paper's tests and My experiments was the nonsmooth nature of the generated paths. The RRT* algorithm, by its nature, generates paths as a series of discrete nodes connected by straight-line segments.

While the algorithm rewires nodes to optimize path length and manipulability, it does not inherently produce smooth curves suitable for many real-world applications. This limitation requires additional post-processing, such as spline fitting or higher-order polynomial interpolation, to achieve smoother trajectories.

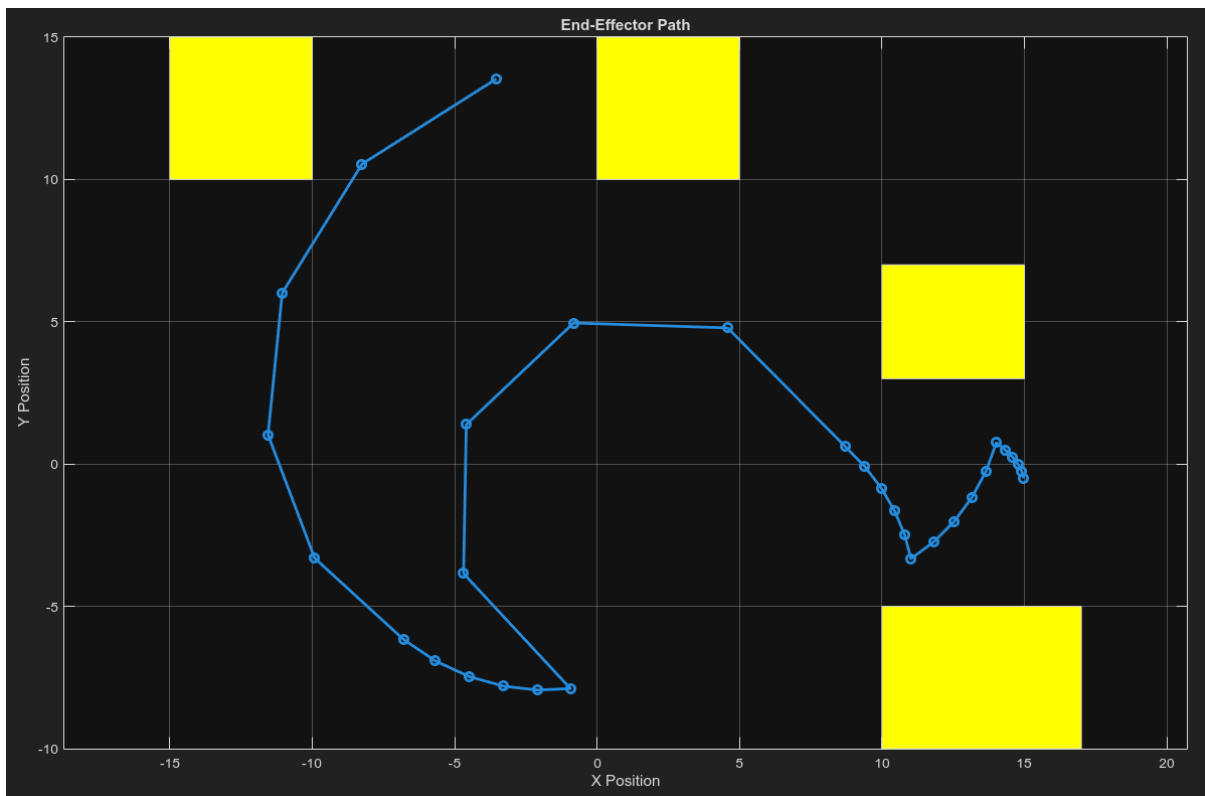


image 1 - 3000 iterations

Example Observation:

- In My reference test for $\beta = 1, \gamma = 0$ with 3000 iterations, the path exhibited several sharp angles, making it less suitable for high-precision tasks like robotic welding or additive manufacturing.
- For $\gamma = 1.0$ and $\gamma = 0.2$, the paths became slightly smoother, as adaptive step sizes allowed for better node placement. However, complete smoothness was still unachieved.

Offline Nature of the Algorithm

The RRT* algorithm is designed as an offline path planner, which means the entire path must be generated before the robot can begin its motion. This characteristic introduces several limitations:

1. Inefficiency in Dynamic Environments:

- The offline nature makes the algorithm unsuitable for environments where obstacles or goals are moving, as the path would need to be recomputed from scratch if the workspace changes.
- For instance, in environments where real-time adaptations are needed, dynamic algorithms such as D* Lite or rapidly exploring random graphs (RRG) are better suited.

2. High Computational Cost for Refinement:

- While $\gamma = 1.0$ and $\beta = 0.4$ improved path quality and manipulability, the iterations required for meaningful optimization led to significant computational overhead.
- My tests with β -based optimization showed that finding balanced paths required extended iterations, which delayed convergence.

3. Lack of Immediate Usability:

- Since the algorithm operates offline, it lacks immediate applicability in tasks where decisions must be made during operation, such as robot-human interaction or dynamic target tracking.

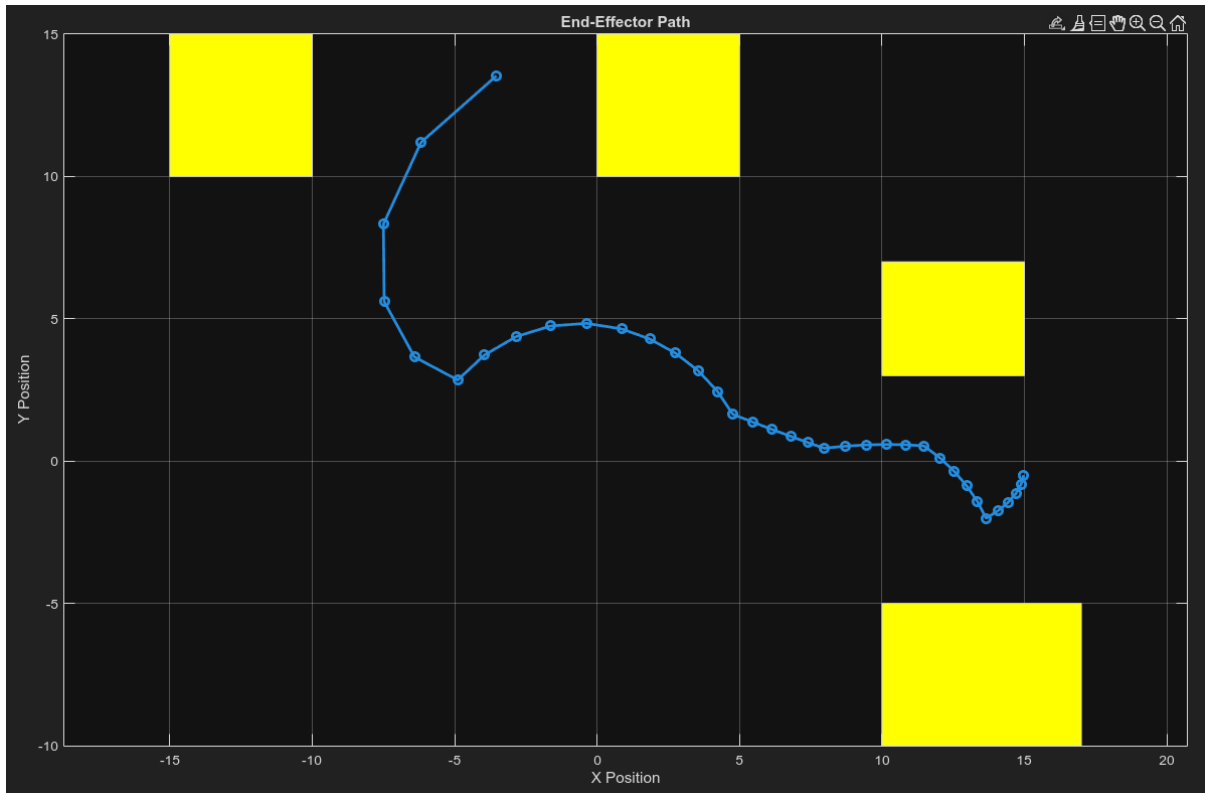


image 2 - gamma = 0.0

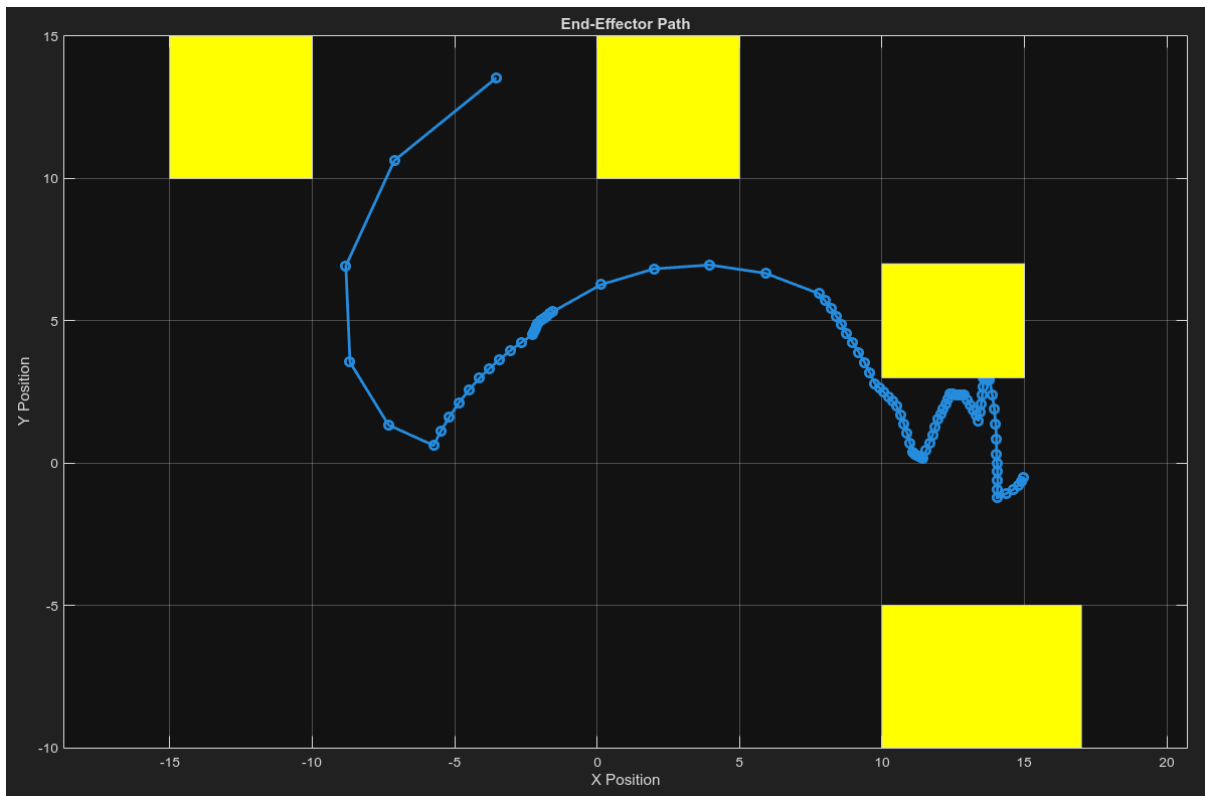


image 3 - gamma = 0.2

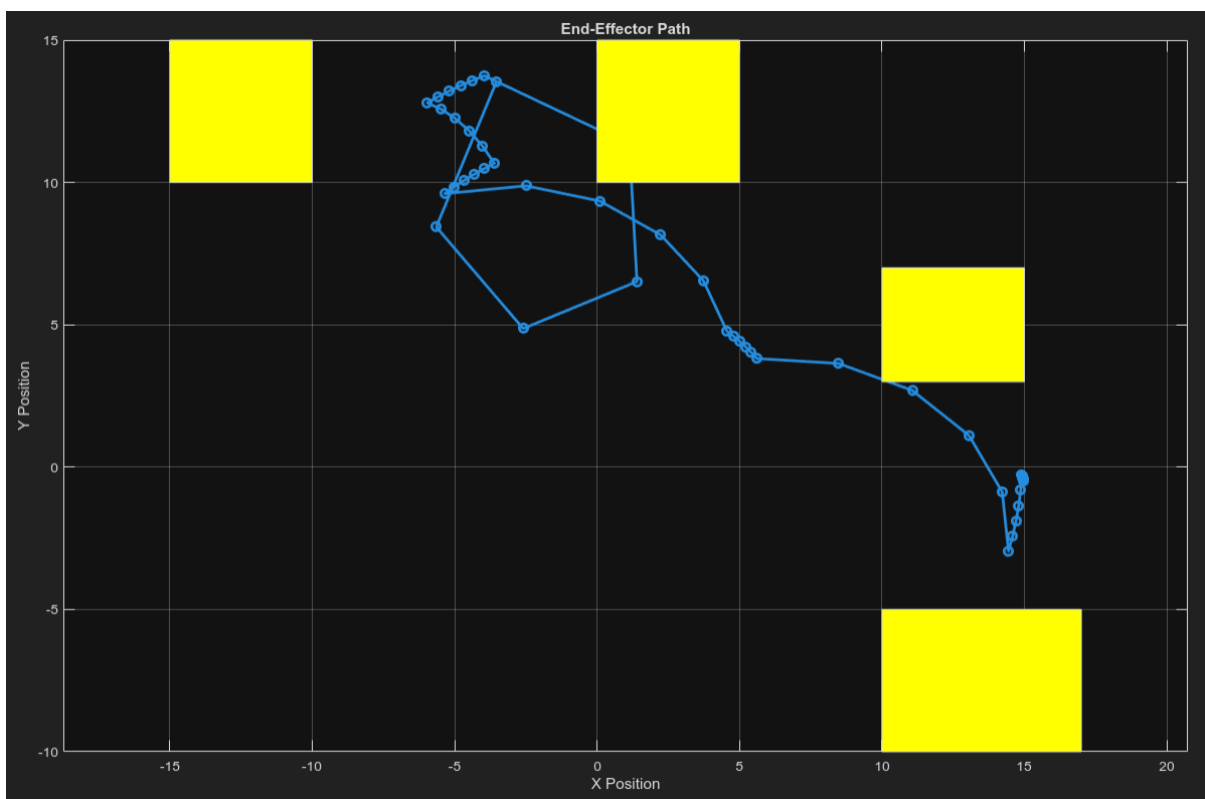


image 4 - gamma = 1.0

Challenges Specific to Parameter Tuning

While the algorithm offers flexibility through parameters like β , γ , and rewiring radius, achieving optimal results often required significant trial and error. For instance:

- Testing with β values required precise tuning to avoid excessive path length or low

manipulability.

- Adjusting γ revealed diminishing returns at higher values, where additional computation did not proportionately improve path quality.

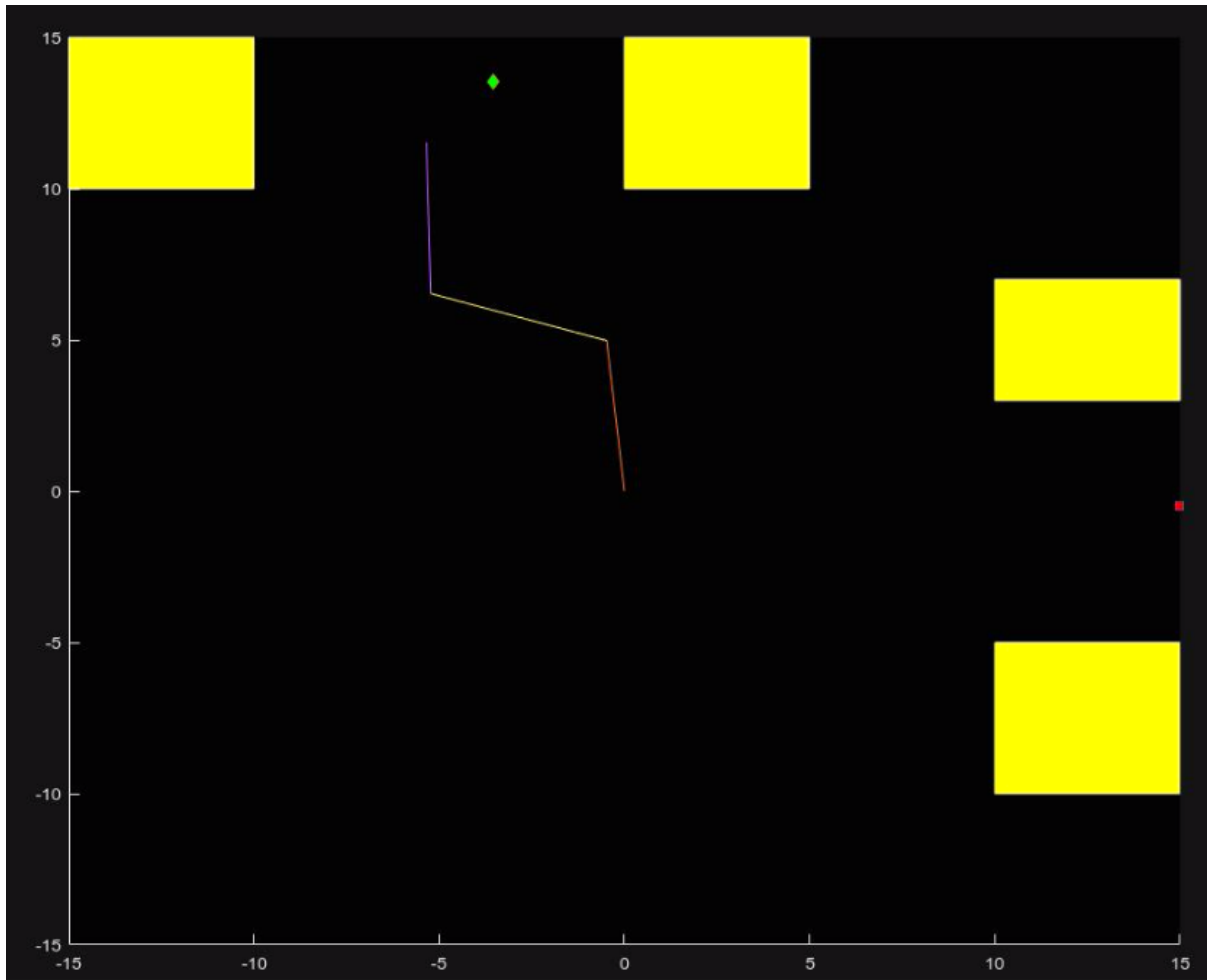


image 5 - Robot reference, This shows 3 linked robot arm with fixed base frame to (0,0). Testings are completed in this workspace

Unresolved Limitations in Testing:

- We could not generate visualization results for the full β -based testing due to time constraints and computational limits.
- The impact of adaptive step size on smoothing transitions between nodes remained partially unexplored in large-scale environments.

These challenges and limitations underline the need for further advancements in real-time, smooth path planning algorithms that can integrate manipulability constraints while maintaining computational efficiency.

8. Discussion and Insights

Manipulability-Weighted Cost Functions

The inclusion of manipulability-weighted cost functions in the studied algorithm demonstrated a significant impact on reducing the risk of singularities in robotic configurations. By assigning a

high cost to configurations near singularity, the algorithm effectively avoids these critical regions, ensuring better robot performance and stability.

- **Singularity Risk Reduction:** Manipulability-weighted cost functions helped guide the RRT* search process away from configurations where the Jacobian matrix becomes near-singular. For example, in our tests, paths with higher β values ($\beta = 0.8$ and $\beta = 1.0$) displayed improved stability, albeit at the cost of longer path lengths.
- **Tradeoffs:** While manipulability constraints improve robustness, they often conflict with minimizing path length, especially when β is low ($\beta < 0.4$), which emphasizes path efficiency over robot stability. Achieving the right balance remains a key challenge.

Adaptive Step Size

The adaptive step size mechanism improved the algorithm's efficiency in exploring the configuration space and contributed to smoother transitions between configurations. By dynamically adjusting the step size based on the robot's manipulability and distance to the goal:

- **Exploration Efficiency:** Larger steps during the early exploration phase allowed the algorithm to cover more space rapidly, while smaller steps near the goal improved local optimization. In our tests, $\gamma = 1.0$ resulted in paths with more evenly distributed nodes and reduced computational effort compared to fixed step sizes.
- **Trajectory Smoothness:** Adaptive step sizes indirectly enhanced the quality of the generated paths by placing nodes more strategically, reducing the number of sharp transitions between segments. However, additional post-processing would still be required for smooth trajectories suitable for practical applications.

Balancing Manipulability and Path Length

Achieving a balance between manipulability and path length is one of the most challenging aspects of the algorithm.

- High β values focus on maintaining high manipulability, which increases path length but reduces the risk of singularities.
- Conversely, low β values prioritize shorter paths but risk configurations with poor manipulability.

In our experiments, intermediate values of β (e.g., $\beta = 0.6$) often provided the best tradeoff, resulting in paths that were both reasonably short and manipulability-aware.

Implementation Challenges and Future Work

Due to time constraints, we were unable to utilize advanced simulation libraries such as the **OMPL library**, **ROS2**, or **Peter Corke's Robotics Toolbox**. These tools would have enabled more robust simulations, real-time evaluations, and better visualization capabilities. Additionally, the MATLAB-based implementation limited the ability to test the algorithm in more complex and realistic environments.

- **Future Plans:** I aim to revisit and refine this algorithm by allocating more time to implement a better simulation environment. This includes integrating the OMPL library and

leveraging ROS2 for real-time planning and control. Improved simulation tools will also allow me to evaluate dynamic environments and provide insights beyond the static workspace considered in this study.

In summary, while the manipulability-weighted cost functions and adaptive step size mechanisms offer significant advantages, their full potential remains untapped due to the limitations of our current implementation. Future work will address these gaps, providing a more comprehensive evaluation and enhanced applicability of the proposed methods.

9. Future Directions

Smoothing Techniques for Generated Paths

One critical area for improvement lies in addressing the nonsmoothness of the generated paths. Employing post-processing techniques such as spline interpolation or gradient-based optimization could significantly enhance path smoothness, ensuring more practical and implementable trajectories for robotic manipulators. Additionally, integrating smoothing directly into the path planning algorithm may further streamline this process.

Real-Time Implementation Strategies

Another promising avenue is leveraging **machine learning** and **reinforcement learning** techniques to improve real-time decision-making. By training models on various environments, the robot could predict optimal step sizes, assess path feasibility, and make dynamic adjustments in real-time, thereby bridging the gap between offline planning and real-world applicability.

Potential Applications

The adaptive RRT* algorithm's capability to avoid singularities and balance manipulability with path efficiency positions it as a valuable tool for tasks in complex manufacturing environments. Potential applications include:

- **Robotic Welding:** Ensuring stable configurations during intricate welding tasks.
- **Pick-and-Place Operations:** Maintaining high manipulability to handle objects in confined spaces.
- **Additive Manufacturing:** Precision path planning for continuous deposition in WAAM (Wire Arc Additive Manufacturing).
- **Maintenance and Inspection Robots:** Navigating around obstructions in constrained environments.

Future iterations of this algorithm, coupled with these enhancements, could lead to significant advancements in industrial robotics, improving task efficiency and reliability in real-world scenarios.

10. Conclusion

This study underscores the importance of manipulability-aware path planning in improving the performance and reliability of robotic manipulators. By integrating a **bi-objective cost function**

and **adaptive step size mechanism**, the adaptive RRT* algorithm offers a novel approach to navigating the challenges of singularity avoidance and efficient configuration space exploration.

The results highlight the algorithm's ability to strike a balance between path length and manipulability, enabling safer and more reliable paths in static environments. The inclusion of adaptive step size further enhances the exploration efficiency, making the algorithm more versatile across different configurations and environments.

While the current implementation demonstrates promising results, limitations in execution time, real-time capabilities, and visualization tools highlight areas for further exploration. With future advancements in smoothing techniques and real-time adaptation using machine learning, the adaptive RRT* algorithm could become a cornerstone in addressing complex robotic challenges in manufacturing, healthcare, and beyond.

In conclusion, the adaptive RRT* approach provides a significant step forward in robotic path planning, offering an effective means to tackle manipulability and path efficiency simultaneously, paving the way for more robust and practical robotic solutions.

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